A Lorentz-Scale Partial Result for Lempert's Problem

Automatic Math Discovery Smoke Test

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Source and Target

The source paper is Andreas Defant, Daniel Galicer, Martin Mansilla, Mieczyslaw Mastylo, and Santiago Muro, *Local constants and Bohr's phenomenon for Banach spaces of analytic polynomials*, arXiv:2509.26326.

After Theorem 2.21, on page 28 of the arXiv PDF, the authors recall Lempert's problem: whether the projection constants in Theorem 2.21 can be replaced by the unconditional basis constants of the monomial basis. More concretely, for a Banach sequence lattice X, the open question is whether

$$X \equiv \ell_1 \quad \Longleftrightarrow \quad \sup_{m,n} \chi_{\mathrm{mon}}(\mathcal{P}_m(X_n))^{1/m} < \infty.$$

They also recall a theorem of Bayart which gives the following necessary condition for the right-hand side: for every $\varepsilon > 0$, there is $D(\varepsilon) > 0$ such that

$$\|\mathrm{id} : X_n \to \ell_1^n\| \le D(\varepsilon)(\log\log n)^{\varepsilon} \qquad (n \ge 3).$$

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Determining whether, in Theorem 2.21, the projection constant of $\mathscr{P}_m(X_n)$ can be replaced by the unconditional basis constant of the monomial basis remains an open problem, known as 'Lempert's problem' in [26]. Specifically, it is unknown if $X \equiv \ell_1$ if and only if $\sup_{m,n} \boldsymbol{\chi}_{\mathbf{mon}}(\mathscr{P}_m(X_n))^{\frac{1}{m}} < \infty$.

We recall a significant partial solution provided in [3] (see also [26, Theorem 20.26]), which essentially shows that in this case X is very similar to ℓ_1: there is $D = D(\varepsilon) > 0$ such that $\|\mathrm{id}\colon X_n \to \ell_1^n\| \le D(\log\log n)^{\varepsilon}$ for all n.

As we have observed, the projection constant and the unconditional basis constant are closely related. This indicates that, in a certain sense, the results of Theorem 2.21 can be viewed as a solution to a weaker version of Lempert's problem.

Result

We settle this question on the classical Lorentz scale.

Theorem 1. *Let $X = \ell_{p,q}$, where $1 \le p < \infty$ and $1 \le q \le \infty$, with the usual Lorentz sequence norm. If*

$$\sup_{m,n} \chi_{\mathrm{mon}}(\mathcal{P}_m(X_n))^{1/m} < \infty,$$

then $p = q = 1$. Hence, among Lorentz sequence spaces, Lempert's condition can hold only for $X = \ell_1$. Conversely, $X = \ell_{1,1} = \ell_1$ is the known positive model in Lempert's problem.

Proof Intuition

Bayart's theorem says that any counterexample to Lempert's expected classification must look almost like ℓ_1 on finite sections: the identity $X_n \to \ell_1^n$ may grow, but no faster than every tiny power of $\log\log n$. Lorentz spaces miss this threshold in a very visible way. For $p > 1$, the all-ones vector already gives polynomial growth $n^{1/p'}$. For $p = 1$ but $q > 1$, the harmonic vector $(1, 1/2, \ldots, 1/n)$ gives a positive power of $\log n$. These lower bounds are far too large to be compatible with Bayart's necessary condition.

Proof

Assume

$$\sup_{m,n} \chi_{\mathrm{mon}}(\mathcal{P}_m(X_n))^{1/m} < \infty.$$

By the Bayart theorem recalled in the source paper, for every $\varepsilon > 0$ there is $D(\varepsilon) > 0$ such that

$$\|\mathrm{id} : X_n \to \ell_1^n\| \leq D(\varepsilon)(\log\log n)^\varepsilon \qquad (n \geq 3). \tag{1}$$

We show that (1) fails for every Lorentz space except $\ell_{1,1}$.

First suppose $p > 1$. Let

$$u^{(n)} = (1, \ldots, 1, 0, \ldots)$$

with n nonzero coordinates. The Lorentz fundamental function satisfies

$$\|u^{(n)}\|_{\ell_{p,q}} \asymp n^{1/p},$$

with constants depending only on p, q, also when $q = \infty$. Since $\|u^{(n)}\|_1 = n$, it follows that

$$\|\mathrm{id} : \ell_{p,q}^n \to \ell_1^n\| \geq c_{p,q} n^{1-1/p} = c_{p,q} n^{1/p'}.$$

This contradicts (1) for any fixed $\varepsilon > 0$, once n is large.

It remains to consider $p = 1$. If $q = 1$, then $\ell_{1,1} = \ell_1$. Let $1 < q < \infty$, and set

$$h^{(n)} = (1, 1/2, \ldots, 1/n, 0, \ldots).$$

For the usual Lorentz norm,

$$\|h^{(n)}\|_{\ell_{1,q}} = \left(\sum_{k=1}^n (1/k)^q k^{q-1}\right)^{1/q} = \left(\sum_{k=1}^n \frac{1}{k}\right)^{1/q} \asymp (\log n)^{1/q}.$$

At the same time,

$$\|h^{(n)}\|_1 = \sum_{k=1}^n \frac{1}{k} \asymp \log n.$$

Thus

$$\|\mathrm{id} : \ell_{1,q}^n \to \ell_1^n\| \geq c_q (\log n)^{1-1/q} = c_q (\log n)^{1/q'}.$$

Choosing, for instance, $0 < \varepsilon < 1/q'$, this again contradicts (1) for large n.

Finally, if $p = 1$ and $q = \infty$, the same vector $h^{(n)}$ satisfies

$$\|h^{(n)}\|_{\ell_{1,\infty}} = \sup_{1 \le k \le n} k \cdot \frac{1}{k} = 1, \qquad \|h^{(n)}\|_1 \asymp \log n.$$

Therefore

$$\|\mathrm{id} : \ell_{1,\infty}^n \to \ell_1^n\| \ge c \log n,$$

contradicting (1).

Hence the bounded-root monomial unconditionality condition cannot hold for any Lorentz sequence space $\ell_{p,q}$ except $\ell_{1,1} = \ell_1$, as claimed.

Verification Notes

The proof uses only two inputs beyond the source definitions:

- Bayart's necessary condition, quoted in the source paper after Theorem 2.21 and attributed there to *Maximum modulus of random polynomials*.
- The elementary Lorentz norm computations $\|(1, \dots, 1)\|_{\ell_{p,q}^n} \asymp n^{1/p}$ and $\|(1, 1/2, \dots, 1/n)\|_{\ell_{1,q}^n} \asymp (\log n)^{1/q}$ for $1 < q < \infty$, with the weak-ℓ_1 norm equal to 1 in the case $q = \infty$.

No numerical computation is used.

Scope and Human Review Recommendation

This is a partial result only. It does not solve Lempert's problem outside the Lorentz family, and it does not improve Bayart's general necessary condition. It records that the Lorentz spaces naturally present in the source paper cannot supply a counterexample to Lempert's expected classification.

Human review should check the exact formulation of the Lorentz norm convention used for $q = \infty$, the dependence of the constants on p, q, and whether the source's word "$X \equiv \ell_1$" is intended as isometric equality of the sequence lattice or Banach-space isomorphism. Under the standard Lorentz normalizations, the finite-section identity lower bounds are invariant up to absolute constants and the conclusion is unchanged.

Search Notes

Run indexes were searched for arXiv:2509.26326, Lempert, monomial basis, Bayart, $\log \log$, and Lorentz variants. External searches on 2026-06-21 for "Lempert's problem monomial basis", "Lempert's problem log log Bayart", and "Lempert Lorentz monomial" found no later full solution or duplicate Lorentz-scale packet.

References

- A. Defant, D. Galicer, M. Mansilla, M. Mastylo, and S. Muro, *Local constants and Bohr's phenomenon for Banach spaces of analytic polynomials*, arXiv:2509.26326.
- F. Bayart, *Maximum modulus of random polynomials*, Q. J. Math. 63 (2012), no. 1, 21–39.

- J. Lindenstrauss and L. Tzafriri, *Classical Banach Spaces I: Sequence Spaces*, Springer, 1977, for standard Lorentz sequence-space notation and finite-section estimates.